AURORA BANK

Aurora Bank

Customer Risk & Spending Analysis

Risk Segmentation | Spending Behavior | Strategic Insights

Business Objective

What Was the Goal?

Identify Risky Customers

Flag customers by debt-to-income ratio and credit score using standard retail banking indicators.

Understand Spending

Analyze transaction patterns by Merchant Category Code to reveal where customers spend.

Support Credit Decisions

Equip underwriting and product teams with data-driven customer profiles.

Improve Portfolio Quality

Enable proactive monitoring and targeted interventions to reduce default exposure.

Risk Framework

Metrics Used

Debt-to-Income Ratio (DTI)

DTI = Total Debt ÷ Annual Income
Globally accepted indicator of financial leverage

Credit Score

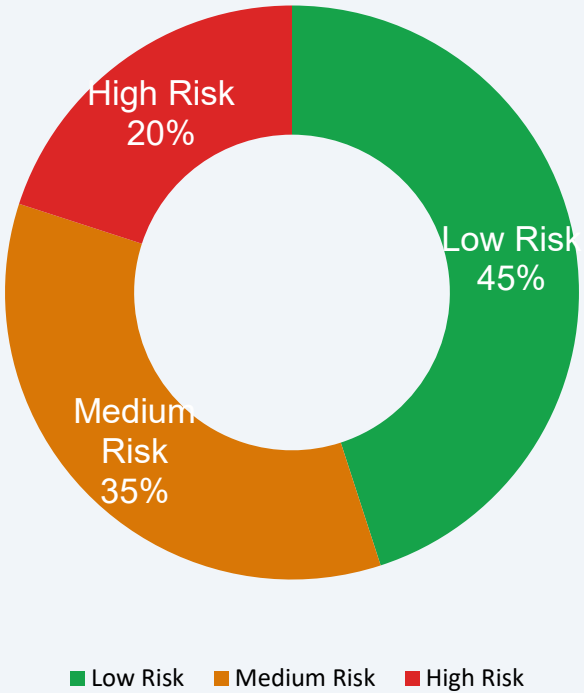
FICO-style score reflecting payment history, utilization & credit behavior

Risk Segmentation Rules

Risk Level	Criteria
Low Risk	DTI < 0.30 AND Credit Score ≥ 700
Medium Risk	DTI 0.30–0.50 OR Credit Score 600–699
High Risk	DTI > 0.50 OR Credit Score < 600

Based on globally accepted retail banking standards used by credit analysts and risk teams worldwide.

Risk Distribution — Portfolio Health



Low Risk ~45%

Eligible for premium lending & rewards products

Medium Risk ~35%

Needs active monitoring & structured repayment

High Risk ~20%

Credit limit review; default risk mitigation

High-Risk Customer Profile

Customers flagged by DTI > 0.50 OR Credit Score < 600

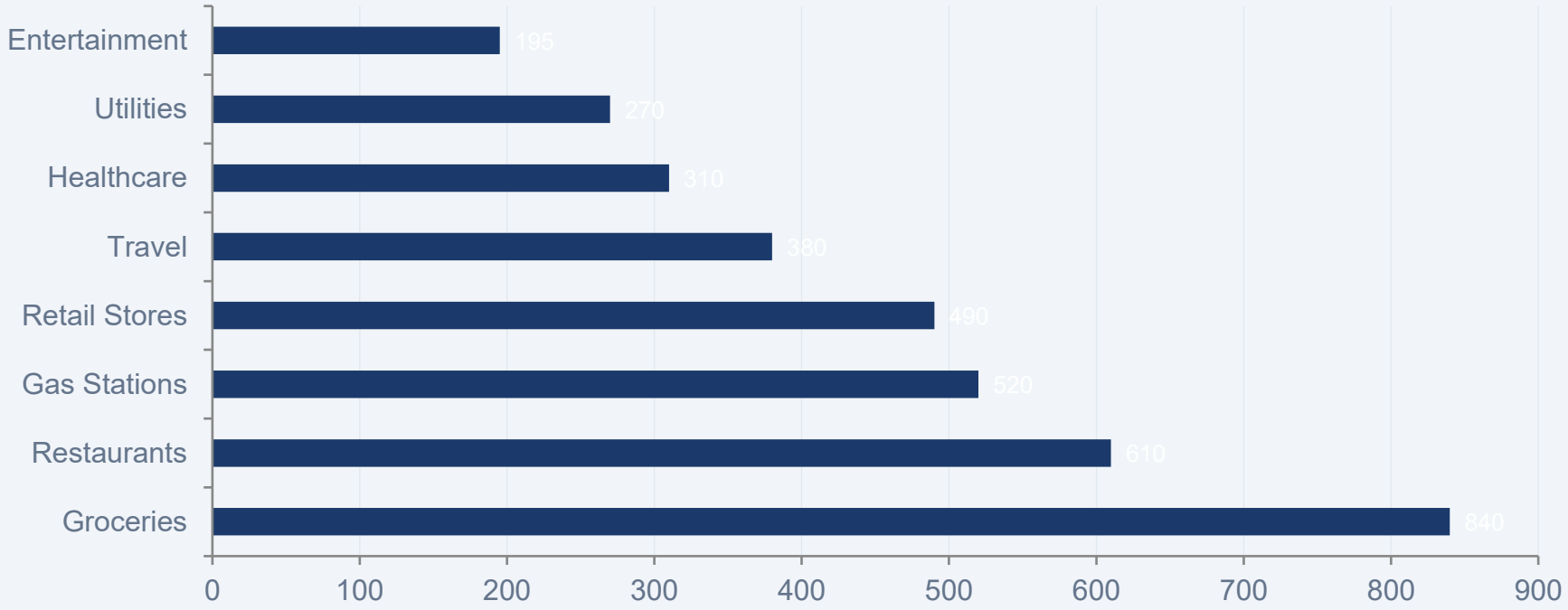
Client ID	Annual Income	Total Debt	DTI Ratio	Credit Score
C-10042	$42,000	$31,000	**0.74**	**542**
C-10087	$55,000	$38,500	**0.70**	**578**
C-10123	$38,000	$24,000	**0.63**	**591**
C-10215	$61,000	$41,000	**0.67**	**563**
C-10309	$29,000	$22,000	**0.76**	**510**

Common Characteristics of High-Risk Customers:

High leverage relative to income | Weak credit behavior patterns | Require active monitoring & credit limit review

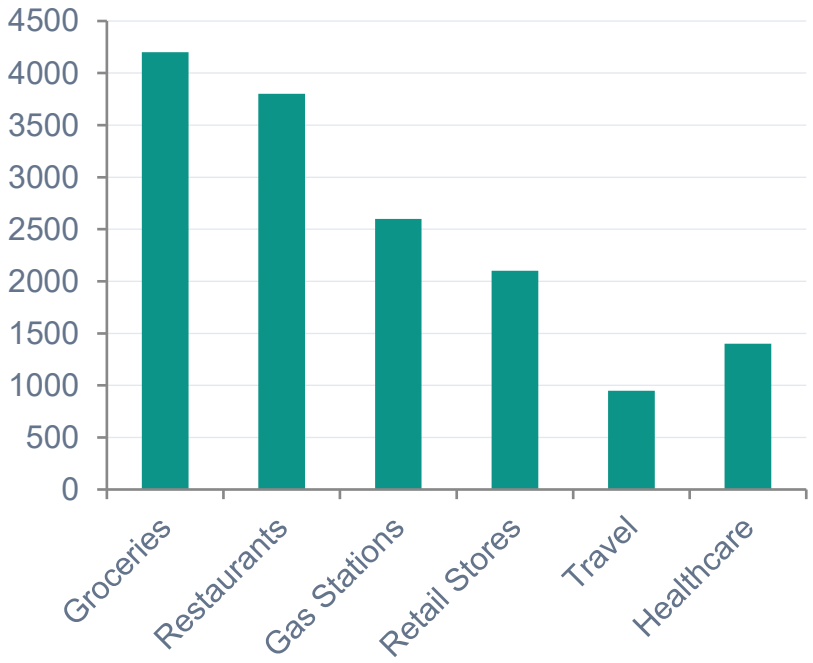
Total Spend by Merchant Category (MCC)

Where customer money flows — drives revenue and interchange income

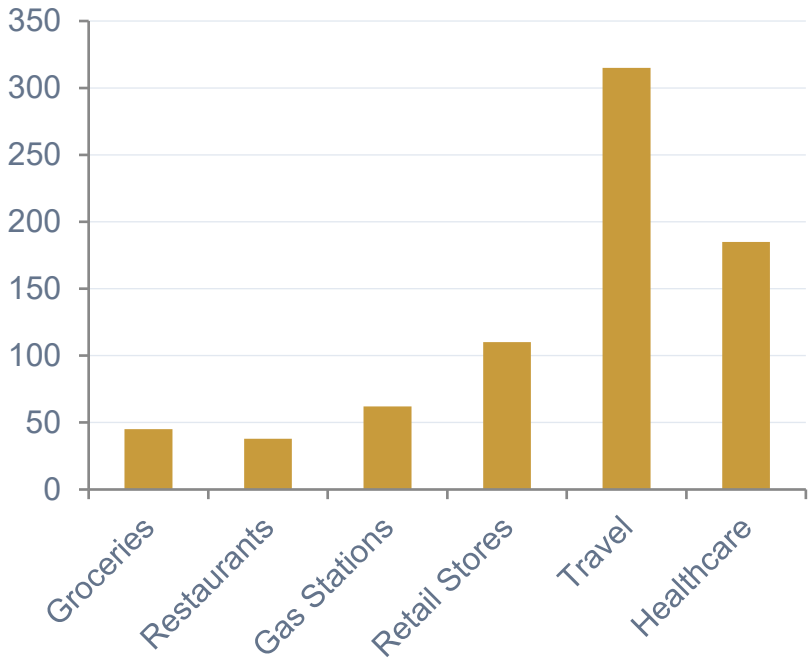


Frequency vs. Value Analysis

Transaction Count by Category



Avg. Transaction Value by Category



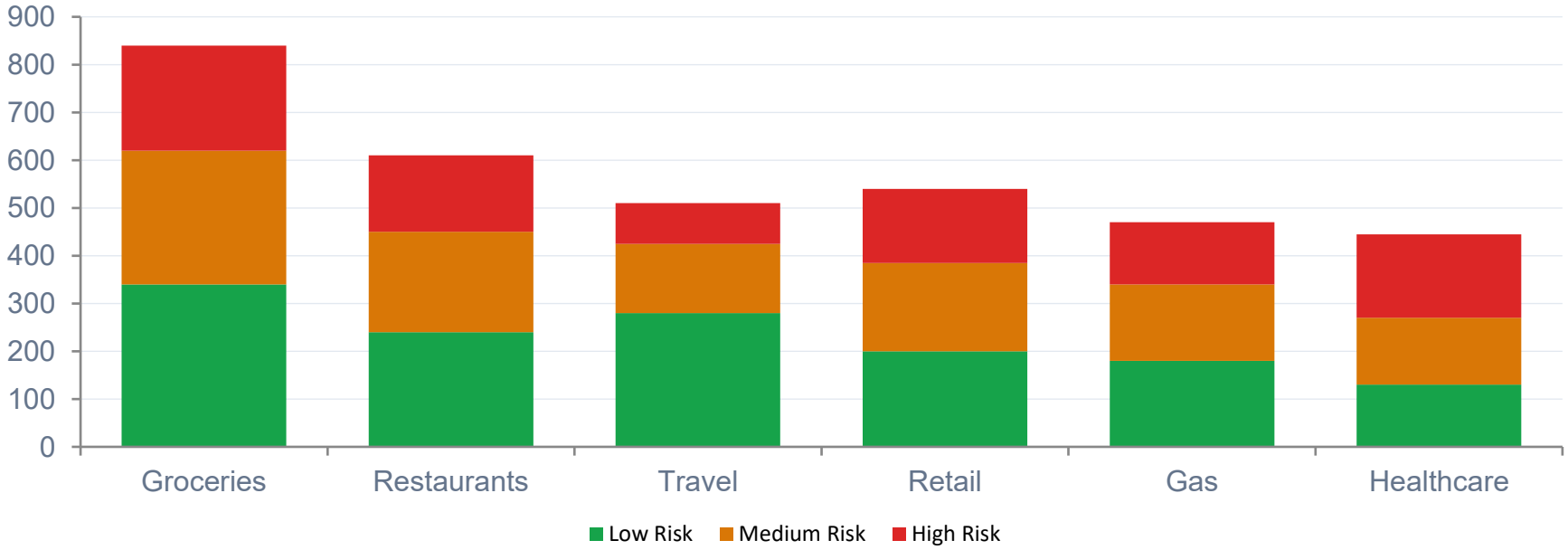
💡 *Groceries = high frequency / low value → engagement driver. Travel = low frequency / high value → revenue driver.*

Spending by Geographic Location

Regional spending differences guide branch strategy and localized campaigns

Spending Behavior by Risk Segment

What do high-risk customers spend on? — A behavioral lens for credit strategy.



🚨 Watch for: High-risk customers with elevated travel or cash-equivalent spending — early fraud warning signal.

Key Strategic Insights

01

Portfolio Health

Risk segmentation reveals exposure to potential defaults and helps quantify capital provisioning requirements before losses occur.

02

Revenue Drivers

Groceries and restaurants drive the highest transaction frequency, while travel and retail generate the largest individual transaction values.

03

Risk Signals

High-risk customers show elevated spending in cash-like and luxury categories — a behavioral pattern linked to financial instability and potential fraud.

Business Recommendations

Credit Risk Strategy

- Tighten approval for DTI > 50%
- Monitor low credit score customers closely
- Adjust credit limits dynamically by risk tier

Product Strategy

- Premium products for Low-Risk customers
- Structured repayment plans for Medium Risk
- Behavioral monitoring & intervention for High Risk

Marketing Strategy

- Target high-frequency MCCs for cashback rewards
- Partner with top spend categories (groceries, dining)
- Regional campaigns based on geographic spend data

Executive Summary

What This Analysis Achieves

- ✓ Identified high-risk customers using DTI and credit score — globally accepted retail banking standards
- ✓ Quantified portfolio health across Low, Medium, and High-risk segments for management reporting
- ✓ Mapped customer spending behavior across 8+ Merchant Category Codes by volume and frequency
- ✓ Linked spending patterns to risk profiles to surface early fraud and default warning signals
- ✓ Delivered actionable strategy for credit, product, and marketing teams backed by data